

Ahnaf Murshid

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EDUCATION

University of Houston

M.S. in Engineering Data Science and AI

Houston, TX

Aug 2026 – Present

University of Houston

B.S. in Computer Science, Minor in Mathematics

Houston, TX

Aug 2024 – Aug 2026

Texas Tech University

Computer Science coursework – transferred to UH

Lubbock, TX

Aug 2022 – May 2024

EXPERIENCE

University of Houston–Clear Lake

Undergraduate Researcher, Robust Encrypted Traffic Classification

Houston, TX

Jun 2026 – Sep 2026

- First-authored a paper accepted at **IEEE CCSS 2026** on robust transformer-based traffic classification
- Diagnosed shortcut learning in a **132M-parameter** BERT model: corrupting 16 header bytes cut macro-F1 from **0.98 to 0.35**
- Built a training-time regularizer retaining **0.85 vs. 0.53 macro-F1** under full header loss, with **5× lower** seed variance
- Improved robustness to black-box and white-box **adversarial attacks** by **10–12 macro-F1 points** at under 0.4% training cost
- Ran **15 fine-tuning experiments** with multi-seed runs, ablations, and bootstrap CIs; caught an unstable metric pre-publication

University of Houston

Research Assistant, SZ3 Scientific Data Compression

Houston, TX

Jan 2025 – May 2025

- Achieved a **7× compression ratio** at **97% data accuracy** across 7+ large-scale weather and seismic datasets
- Built automated testing pipelines benchmarking compression algorithms on HPC-scale data with PSNR and L2-norm metrics

Texas Tech University

Student Assistant

Lubbock, TX

Jun 2023 – Jul 2024

- Managed academic records for a **500+ student** department at 100% data integrity across 2+ semesters
- Cut record retrieval time by restructuring and cleaning large datasets in Excel

PROJECTS

Autonomous Rover – Camera-Only Localization

Visual Place Recognition, Deep Learning

PyTorch, DINOv2 ViT, TorchScript, ROS 2

- Built a camera-only localizer – one frame in, map position out – at **0.25 m median error**, 90% within 1 m
- Fine-tuned a **DINOv2 vision transformer** on a self-collected **13,875-image** day/night dataset
- Cut error **62%** under unfamiliar lighting (0.71 m → 0.27 m) after tracing four failed runs to a misleading validation split
- Deployed to a Raspberry Pi, where one camera frame corrected a **5.66 m** position error in **3 seconds**
- Added natural-language control: an **LLM tool-calling agent** with Whisper speech-to-text issuing navigation goals

Lung Cancer CT Scan Classification Pipeline

Medical Imaging / Deep Learning

PyTorch, OpenCV, scikit-learn

- Classified 5 lung CT categories at **81% test accuracy** with a from-scratch PyTorch CNN trained on **1,535 images**
- **Per-class error analysis** exposed what accuracy hid: benign, the rarest class, at 0% recall on all 18 test cases
- Automated preprocessing for mixed JPG/PNG/RGBA sources: RGBA compositing, 224×224 resizing, stratified 70/15/15 split
- Identified class weighting and a pretrained backbone as the fix, with **macro-F1** replacing accuracy as the metric

TECHNICAL SKILLS

Languages: Python, SQL, C++, C, Java, R, JavaScript/TypeScript

ML & AI: PyTorch, scikit-learn, Transformers (BERT, DINOv2 ViT), CNNs, transfer learning & fine-tuning, metric learning, data augmentation, adversarial robustness, model evaluation & experiment design, LLM tool calling

Data & Vision: Pandas, NumPy, OpenCV, Matplotlib, seaborn, statistical analysis (bootstrap confidence intervals)

Tools: Git, Linux, Docker, Kubernetes, AWS, Azure, Flask, ROS 2, TorchScript, Raspberry Pi